

Methylation, Homocysteine and Choline

Folate cycle (MTHFR, MTHFD1, SHMT1), methionine cycle (MTR, MTRR, MAT1A, GNMT), transsulfurization (CBS), choline and betaine pathway (BHMT, PEMT), vitamin B12 (TCN2, FUT2, CUBN, CLYBL, MMUT) and COMT

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Variants by theme	6
5. Genetic score of homocysteine (4 loci)	9
6. Favourable	10
7. Unfavourable	11
8. Mechanisms of ageing	12
9. Practical suggestions	13
10. Limitations	14
11. Methods and sources	15
12. Annex: all variants analysed	16

1. Summary

What was seen. 27 SNPs were read in 20 genes (26 of chip, 1 imputed) and 1 polygenic score, plus a homocysteine score of 4 loci.

MTHFR compound heterozygous. Sérgio F has **C677T** (rs1801133 G/A) and **A1298C** (rs1801131 T/G), almost always in trans, which leaves about 50–60% of the activity of MTHFR. It is not homozygous C677T and does not have R594Q. It is the genetic reason for folate to always be 5-MTHF, which circumvents the step of MTHFR.

Genetically higher homocysteine. Three loci GWAS push it up: **DPEP1 rs154657 A/A**, **MMUT rs9369898 A/A** and **GNMT rs9296404 T/T** (imputed, DR2 1.00; rises homocysteine after a methionine load). The 4 loci score is +0.20 DP above the average, the 92nd percentile in the reference population, which is equivalent to about +0.6–0.8 µmol/L: a real but small effect. In the opposite direction, **CBS** has 2 protective alleles (rs234706 A and rs234709 T) and does not have the homocystinuria variant.

B12 without relevant genetic risk. MTR and MTRR with common genotypes, CUBN and CLYBL without the low B12 alleles, FUT2 heterozygous secretor; **TCN2 776G** (1 copy) gives a little less active B12. Polygenic B12 score in p34,0 (typical).

Hill and methyl intake. **PEMT Val175Met** and **BHMT Arg239Gln** heterozygous (need for slightly larger choline); **COMT Val/Val** (haplotype LPS, high activity: more consumption of SAM); **CPS1 C/C** (genetically lower glycine).

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold, or outside the imputed zone, SNP is treated as SNP not found in the data.
- **Variants:** 27 SNPs: 26 of the chip, 1 imputed and 0 not found. All genotypes were reread directly in the raw DNA and in the imputation for this report.
- **Strand:** All genotypes are in the GRCh38 strand +. 0 SNPs needed conversion to complementary strand and 2 are palindromic (A/T or C/G). When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Chromosome X:** Sérgio F has only one copy of X outside the pseudoautosomal regions, so the SNPs of X appear with a single letter (hemizotic).
- **Polygenic scores (PGS):** 1 PGS Catalog scores calculated locally with 1.082.449 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$), calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVABLE/ALTO; ≤ 20 : MENOS PROVABLE/BAIXO; between 20 and 80: TIPIC.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

Percentage in the reference population. There is no homocysteine score in the catalog with many variants; the 4 loci score described in the next section was used.

Polygenic score

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Vitamin B12 in blood	PGS003158 (Am J Hum Genet 2022)	1.082.449 1.113.819 (weighted coverage 96%)	34,0	TIPIC	typical

4. Variants by theme

For each SNP: genotype (strand +), origin, effect allele and number of copies, judgment and evidence. In the "protective" SNPs, having the allele is favourable; in the "no judgment", the allele does not have a good or bad meaning.

Folate cycle

MTHFR forms the 5-metil-THF (5-MTHF), the methyl donor that recycles homocysteine in methionine. MTHFD1 and SHMT1 feed the cycle with carbon units.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
MTHFR	rs1801133	C677T (Ala222Val)	G/A	chip	A (1 copy/s)	unfavourable (light)	Thermolabile enzyme: ~35% less activity per allele; +0.16 DP homocysteine by allele	Strong
MTHFR	rs1801131	A1298C (Glu429Ala)	T/G	chip	G (1 copy/s)	unfavourable (light)	Small decrease of activity; relevant especially in heterozygous compound with C677T	Moderate (in compost)
MTHFR	rs2274976	G1793A (Arg594Gln)	C/C	chip	T (0 copy/s)	favourable	Rare variant of lower activity	Weak
MTHFD1	rs2236225	G1958A (Arg653Gln)	G/G	chip	A (0 copy/s)	favourable	Lower 10-formil-THF synthesis; greater choline requirement	Moderate
SHMT1	rs1979277	C1420T (Leu474Phe)	G/G	chip	A (0 copy/s)	favourable	Small and inconsistent effect	Weak
SLC19A1 (RFC1)	rs1051266	G80A (His27Arg)	T/C	chip	C (1 copy/s)	without judgment	Reduced folate carrier; inconsistent associations	Weak

Methionine cycle and methyl groups

MTR (with B12) and MTRR recycle homocysteine in methionine; MAT1A forms SAM, the universal methyl donor; GNMT spends the excess of SAM. CPS1 controls glycine.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
MTR	rs1805087	A2756G (Asp919Gly)	A/A	chip	G (0 copy/s)	neutral	The G allele is associated with slightly lower homocysteine (small effect)	Weakness
MTR	rs2275565	intronic (GWAS)	G/G	chip	T (0 copy/s)	neutral	The T allele low to homocysteine (−0.054 DP)	Moderate
MTRR	rs1801394	A66G (Ile22Met)	A/A	chip	G (0 copy/s)	favourable	Lower MTRR affinity for MTR; modest effect	Weakness
MTRR	rs1532268	C524T (Ser175Leu)	C/C	chip	T (0 copy/s)	favourable	Small and inconsistent effect	Weak
MAT1A	rs3851059	intronic	G/A	chip	A (1 copy/s)	unfavourable (light)	Possible lower activity of MAT1A (SAM formation); scarce functional data	Weak
GNMT	rs9296404	promoter	T/T	imputation DR2=1.00	T (2 copy/s)	unfavourable	Homocysteine higher after a methionine load (more methyl group consumption)	Moderate
CPS1	rs1047891	Thr1406Asn	C/C	chip	A (0 copy/s)	neutral	Allele A rises greatly to glycine and lower to homocysteine; C/C : genetically lower glycine	Strong (glycine)

Transsulfurization and clearance of thiols

Homocysteine output for cysteine and glutathione (CBS, B6-dependent) and loci GWAS that modulate homocysteine in the blood (DPEP1, MMUT).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CBS	rs234706	C699T (Tyr233Tyr)	G/A	chip	A (1 copy/s)	favourable	Higher transsulfurization and slightly lower homocysteine	Weakness
CBS	rs234709	intronic (GWAS)	C/T	chip	T (1 copy/s)	favourable	The T allele low to homocysteine (−0.072 DP)	Moderate
CBS	rs5742905	I278T (pathogenic)	A/A	chip	G (0 copy/s)	favourable	Classical homocystinuria (recessive)	Strong (if present)
DPEP1	rs154657	intergenic (GWAS)	A/A	chip	A (2 copy/s)	unfavourable	+0.096 DP of homocysteine by allele	Moderate-strong
MMUT	rs9369898	intergenic (GWAS)	A/A	chip	A (2 copy/s)	unfavourable	+0.045 DP homocysteine by allele	Moderate

Collin and betaine

BHMT recycles homocysteine with betaine (alternative route to folate); PEMT manufactures choline in the liver.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
BHMT	rs3733890	G742A (Arg239Gln)	G/A	chip	A (1 copy/s)	unfavourable (light)	Betaine-choline pathway; effects on inconsistent homocysteine	Weak
PEMT	rs7946	Val175Met	C/T	chip	T (1 copy/s)	unfavourable (light)	Lower activity of PEMT (choline synthesis in the liver); associated with fatty liver in some studies	Weakness
PEMT	rs12325817	−744G>C (promotor)	C/C	chip	G (0 copy/s)	favourable	Increased susceptibility to organic dysfunction with poor choline diet	Weakness

Vitamin B12: absorption and transport

Absorption (FUT2, CUBN), transport (TCN2) and mitochondrial use (CLYBL, MMUT) of vitamin B12.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TCN2	rs1801198	C776G (Pro259Arg)	G/C	chip	G (1 copy/s)	unfavourable (light)	Less holo-transcobalamine (active B12) with the 776G allele; direction depends on the test	Moderate
FUT2	rs601338	W143X (secretor/non-secretor)	G/A	chip	A (1 copy/s)	without judgment	A/A = non-secretory (higher serum B12, different microbiome); heterozygous = secretor	Strong (secretory state)
CUBN	rs1801222	Ser253Phe	G/G	chip	A (0 copy/s)	favourable	More B12 deficiency anemia	Moderate
CLYBL	rs41281112	Arg259Ter	C/C	chip	T (0 copy/s)	favourable	Lower B12 (loss of function of CLYBL)	Moderate

Consumption of methyl groups (COMT)

COMT spends SAM to methylate catecholamines, estrogens and polyphenols.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
COMT	rs4680	Val158Met	G/G	chip	G (2 copy/s)	without judgment	Val (G) = high COMT activity: more consumption of SAM in catecholamine methylation, estrogens and polyphenols	Strong (activity)
COMT	rs6269	promoter (haplotype)	G/G	chip	A (0 copy/s)	without judgment	Haplotype marker LPS/APS/HPS	Moderate

5. Genetic score of homocysteine (4 loci)

Locus	SNP	Allele that rises homocysteine	Sérgio F copies	Allele effect
MTHFR	rs1801133	A (677T)	1	+0,158 DP
DPEP1	rs154657	A	2	+0,096 DP
MTR	rs2275565	G (low T)	2 G (0 T)	−0.054 DP by T
MMUT	rs9369898	A	2	+0,045 DP

Result: +0.20 DP above the mean reference population, the 92nd percentile. In absolute value, about +0.6–0.8 $\mu\text{mol/L}$. CBS rs234709 (1 protective allele, −0.072 DP) and GNMT T/T (effect after methionine loading) do not enter the score and pull in opposite directions.

Interpretation: genetics explains a small part of homocysteine; the rest depends on folate, B12, B2, choline/betaine, diet protein, renal function and drugs.

6. Favourable

- **MTHFR without R594Q** (rs2274976 C/C) and without homozygous C677T.
- **CBS:** 2 Alleles that lower homocysteine (rs234706 A, rs234709 T) and without I278T (homocystinuria).
- **MTRR** A66G and C524T without the effect allele; **MTHFD1** R653Q and **SHMT1** without variant.
- **B12:** CUBN Ser253Phe and CLYBL Arg259Ter absent; typical polygenic score (p34,0).
- **PEMT –744** without the risk allele (rs12325817 C/C at strand +).

7. Unfavourable

- **MTHFR heterozygous compound** C677T + A1298C (~50–60% of activity).
- **Homocysteine p92 score:** DPEP1 rs154657 A/A, MMUT rs9369898 A/A; MTR rs2275565 without the allele.
- **GNMT rs9296404 T/T** (imputed): more homocysteine after a methionine load.
- **Hill:** PEMT Val175Met and BHMT Arg239Gln heterozygous (need slightly higher).
- **TCN2 776G** (1 copy): a little less active B12.
- **MAT1A** Heterozygous rs3851059 (weak).
- **CPS1 C/C:** without the allele that rises glycine (neutral, but removes a protection).

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Epigenetic Changes	MTHFR compound heterozygous and GNMT T/T: less 5-MTHF and more consumption of SAM, the methylation donor of DNA	unfavourable (light)
Epigenetic Derivative	Availability of methyl groups (MTHFR, MTR, MAT1A, COMT Val/Val) determines the maintenance of methylation throughout life	unfavourable (light)
DNA Damage and DNA Repair Deficit	Less 5,10-methylene-THF for thymidine synthesis (MTHFR); MTHFD1 without variant	unfavourable (light)
Vascular Aging	Genetically higher homocysteine (p92 score) damage the endothelium	unfavourable (light)
Oxidative stress	CBS with 2 protective alleles: more transsulfurization for cysteine and glutathione	favourable
Mitochondrial dysfunction	MMUT A/A and CLYBL without loss of function: mitochondrial use of B12	mixed
Metabolic Homeostasis Decline	PEMT Val175Met (hepatic choline, fatty liver) and heterozygous BHMT	unfavourable (light)
Dysbiosis	Heterozygous secretor FUT2: intestinal glycans profile and microbiome	neutral

9. Practical suggestions

All suggestions are suggestion (not applied): the real stack of Sérgio F remains as it is, and any change should be discussed with the doctor.

1. **Folate always as 5-MTHF** (never the synthetic form), as it is already in the stack: circumvents the MTHFR step. *Suggestion (not applied).*
2. **Measure homocysteine with folate in red blood cells and methylmalonic acid (MMA) or holo-transcobalamin**, to separate lack of functional folate from B12 (TCN2 776G). *Suggestion (not applied).*
3. **Hill and betain by feeding** (eggs, fish, legumes, beetroots, spinach, whole grains): PEMT and BHMT heterozygous. TMG is already in the stack. *Suggestion (not applied).*
4. **Riboflavin (B2)** is the cofactor of thermolabile MTHFR; it is already in the stack; it does not need high doses. *Suggestion (not applied).*
5. **Many fiber and green leaf vegetables** (natural folate); protein distributed throughout the day, without large loads of methionine in a single meal (GNMT T/T). *Suggestion (not applied).*
6. **Glycine and serine by feeding** (collagen, legumes, soybean), because CPS1 C/C gives genetically lower glycine. *Suggestion (not applied).*
7. **Methotrexate or nitrous oxide (anesthesia)** only with 5-MTHF support and warning to the compound MTHFR doctor. *Suggestion (not applied).*
8. **Zero alcohol** (already the rule): alcohol reduces folate and methylation. *Suggestion (not applied).*

10. Limitations

- Small and added effects; homocysteine depends much more on diet, vitamins, kidneys and drugs than on these genes.
- The phase (cis/trans) of the compound MTHFR is not read in the chip; the trans is almost universal, but it is not confirmed.
- The homocysteine score uses only 4 loci; there is no score with many variants in the catalog.
- GNMT rs9296404 is imputed (DR2 1.00).
- Conclusions only from genes; do not include blood analysis.

11. Methods and sources

- **Genes and SNPs:** 27 SNPs chosen from the previous analysis of methylation (51 lines), keeping the functional variants and loci GWAS homocysteine, B12 and choline; SNPs without established effect were removed.
- **Genotypes:** Sérgio F chip (GRCh38, strand +), reread for this report; GNMT comes from local imputation (Beagle 5.5, public panel of genomes complete to 30x; DR2 $\geq$ 0.8). No genetic data came out of the analysis computer.
- **Guidance:** Alleles of effect on strand +; in the genes of strand – (MTHFR, CBS, PEMT) the allele of the gene was converted (e.g. C677T = A in strand +). 2 palindromic SNPs were assumed in the strand +.
- **Homocysteine score:** weighted sum of published effects (van Meurs 2013) for 4 loci, compared to the reference population.
- **Comparison with previous analysis (02-10-2026):** The 26 genotypes of the chip are equal; the GNMT imputed agrees with the previous inference.
- **Context Literature:** van Meurs 2013, Paré 2009, Zeisel (coline), Miller 2002 (TCN2).

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Folate cycle	MTHFR	rs1801133	G/A	chip	A	1	strand +	1 copy of risk allele
Folate cycle	MTHFR	rs1801131	T/G	chip	G	1	strand +	1 copy of risk allele
Folate cycle	MTHFR	rs2274976	C/C	chip	T	0	strand +	without the risk allele
Folate cycle	MTHFD1	rs2236225	G/G	chip	A	0	strand +	without the risk allele
Folate cycle	SHMT1	rs1979277	G/G	chip	A	0	strand +	without the risk allele
Folate cycle	SLC19A1 (RFC1)	rs1051266	T/C	chip	C	1	strand +	1 copy of effect allele
Methionine cycle and methyl groups	MTR	rs1805087	A/A	chip	G	0	strand +	without the protective allele (common genotype)
Methionine cycle and methyl groups	MTR	rs2275565	G/G	chip	T	0	strand +	without the protective allele (common genotype)
Methionine cycle and methyl groups	MTRR	rs1801394	A/A	chip	G	0	strand +	without the risk allele
Methionine cycle and methyl groups	MTRR	rs1532268	C/C	chip	T	0	strand +	without the risk allele
Methionine cycle and methyl groups	MAT1A	rs3851059	G/A	chip	A	1	strand +	1 copy of risk allele
Methionine cycle and methyl groups	GNMT	rs9296404	T/T	imputation DR2=1.00	T	2	strand +	2 risk allele copies
Methionine cycle and methyl groups	CPS1	rs1047891	C/C	chip	A	0	strand +	without the protective allele (common genotype)
Transsulfurization and clearance of thiols	CBS	rs234706	G/A	chip	A	1	strand +	1 copy of the protective allele
Transsulfurization and clearance of thiols	CBS	rs234709	C/T	chip	T	1	strand +	1 copy of the protective allele
Transsulfurization and clearance of thiols	CBS	rs5742905	A/A	chip	G	0	homozygous for the reference allele (monomorphic site on chip)	without the risk allele
Transsulfurization and clearance of thiols	DPEP1	rs154657	A/A	chip	A	2	strand +	2 risk allele copies
Transsulfurization and clearance of thiols	MMUT	rs9369898	A/A	chip	A	2	strand +	2 risk allele copies
Collin and betaine	BHMT	rs3733890	G/A	chip	A	1	strand +	1 copy of risk allele
Collin and betaine	PEMT	rs7946	C/T	chip	T	1	strand +	1 copy of risk allele
Collin and betaine	PEMT	rs12325817	C/C	chip	G	0	strand + (palindromic; assumed +)	without the risk allele
Vitamin B12: absorption and transport	TCN2	rs1801198	G/C	chip	G	1	strand + (palindromic; assumed +)	1 copy of risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Vitamin B12: absorption and transport	FUT2	rs601338	G/A	chip	A	1	strand +	1 copy of effect allele
Vitamin B12: absorption and transport	CUBN	rs1801222	G/G	chip	A	0	strand +	without the risk allele
Vitamin B12: absorption and transport	CLYBL	rs41281112	C/C	chip	T	0	strand +	without the risk allele
Consumption of methyl groups (COMT)	COMT	rs4680	G/G	chip	G	2	strand +	Val/Val: high activity COMT (haplotype LPS)
Consumption of methyl groups (COMT)	COMT	rs6269	G/G	chip	A	0	strand +	0 copy of the effect allele